Equity and Inclusion

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Introduction

In this chapter, we consider the equitable sharing of benefits arising from community forest management (CFM) and what the impacts of CFM are on vulnerable groups.

Benefit sharing

We asked FGD groups whether the benefits of CFM were equitably distributed (Figure 1). On the one hand, a substantial majority of responses across the FGD groups asserted that benefits were being shared equitably. The proportion of community female and community youth groups that felt that benefits were not being shared equitably was significantly high than for the CFMG executive and community male FGD groups. Moreover, even among the CFMG executive and community male groups, there were some that agreed that benefits were not being shared equitably.

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = economic_transformation_benefits_equity_across_groups,
    response = case_match(response,
                          "yes" ~ "Yes",
                          "no" ~ "No",
                          .default = NA),
    response = factor(response, levels = c("Yes", "No")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
```

```r
    ),
    levels = c("Youth", "Male", "Female", "Executive")
  )
) |>
drop_na(response) |>
count(interview_type, response) |>
group_by(interview_type) |>
mutate(prop = n / sum(n)) |>
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
  guide  = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

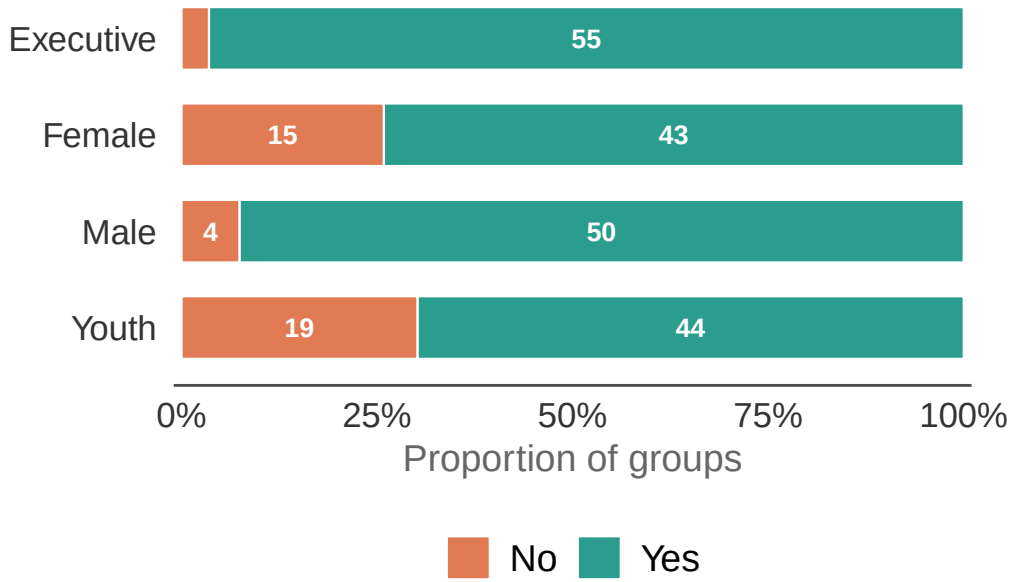

Figure 1: Are the benefits of CFM are equitably shared? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Equity - FGDs - Model methods

To examine differences among FGD types in whether they agreed the benefits of CFM were equitably shared, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    empl = economic_transformation_benefits_equity_across_groups,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interview_type = factor(
      case_match(
```

```r
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M1 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M1      <- emmeans(M1, specs = ~ interview_type, type = "response")
pairwise.M1 <- pairs(emm.M1, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   198.6    219.3    -93.3    186.6      226

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.7383  0.1237  0.2110  0.4329  0.7611

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 1.171e+00 1.082e+00
 province      (Intercept) 5.028e-10 2.242e-05
Number of obs: 232, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)            3.7274     0.8358   4.460 8.21e-06 ***
interview_typeFemale  -2.4647     0.8390  -2.938  0.00331 **
interview_typeMale    -0.7676     0.9178  -0.836  0.40292
```

```
interview_typeYouth   -2.7127     0.8324  -3.259  0.00112 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.888
intrvw_typM -0.724  0.721
intrvw_typY -0.905  0.848  0.719
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

 contrast           odds.ratio     SE  df null z.ratio p.value
 Executive / Female     11.760  9.867 Inf    1   2.938  0.0174
 Executive / Male        2.155  1.977 Inf    1   0.836  0.8371
 Executive / Youth      15.070 12.545 Inf    1   3.259  0.0062
 Female / Male           0.183  0.121 Inf    1  -2.571  0.0498
 Female / Youth          1.281  0.590 Inf    1   0.538  0.9497
 Male / Youth            6.994  4.620 Inf    1   2.944  0.0171

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether the benefits of CFM were equitably shared. Table 1 shows the percentage reporting this broken down by interviewee role. Again a substantial majority of respondents (83% - 92%) thought that benefits were being shared equitably, and the differences among respondents were not significant. However, there was still minority of respondents that did not think benefits were being equitably shared. This suggests that there are some CFs where the benefit sharing arrangements are not optimal or appropriate arrangements are not being followed.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = economic_transformation_benefits_equity_across_groups
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
```

```r
    "trad_leader"      ~ "Traditional leader",
    "dfo"              ~ "District Forestry Officer",
    "NGO_rep"          ~ "NGO representative"
  ),
  interviewee_role = factor(interviewee_role, levels = c(
    "Community member", "CFMG executive member", "Traditional leader",
    "District Forestry Officer", "NGO representative"
  )),
  empl = case_match(empl,
                    "yes" ~ 1,
                    "no" ~ 0,
                    .default = NA)

) |>
group_by(interviewee_role) |>
summarise(
  n          = n(),
  Equity = mean(empl, na.rm = TRUE),
  .groups    = "drop"
) |>
arrange(interviewee_role) |>
gt() |>
cols_label(
  interviewee_role = "Role",
  n                = "n",
  Equity       = "Agree benefits are equitably shared"
) |>
fmt_percent(
  columns  = Equity,
  decimals = 0
) |>
tab_header(
  title    = "Equitable sharing of benefits, as reported by individual interviewees",
  subtitle = "Values represent the % of positive responses. n = the total number of interv
) |>
opt_align_table_header(align = "left") |>
cols_align(align = "left",   columns = interviewee_role) |>
cols_align(align = "center", columns = c(n, Equity)) |>
tab_style(
  style     = cell_text(align = "center"),
  locations = cells_column_labels(columns = interviewee_role)
) |>
```

```
  sub_missing()
```

Table 1: Percentage of individual interviewees reporting that the benefits of CFM are equitably shared by interviewee role.

Equitable sharing of benefits, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses. A positive response indicates that benefits of CFM are equitably shared.

Role	n	Agree benefits are equitably shared
Community member	31	92%
CFMG executive member	63	86%
Traditional leader	75	83%
District Forestry Officer	56	90%
NGO representative	32	96%

Equity - Individual interviewees - Model methods

To examine differences among individual interviewee roles in whether they agreed the benefits of CFM were equitably shared, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    empl                = economic_transformation_benefits_equity_across_groups,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M2 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
```

```r
              (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M2      <- emmeans(M2, specs = ~ interviewee_role, type = "response")
pairwise.M2 <- pairs(emm.M2, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   167.9    198.5    -75.0    149.9      211

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.1332  0.1954  0.2803  0.3690  0.8220

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.3246   0.5698
 province      (Intercept) 0.6187   0.7866
Number of obs: 220, groups:  province_cfmg, 70; province, 8

Fixed effects:
                                 Estimate Std. Error z value Pr(>|z|)
(Intercept)                       3.44865    1.48873   2.317   0.0205 *
interviewee_rolecfmg_exec_member -0.75287    0.91933  -0.819   0.4128
interviewee_roletrad_leader      -0.93842    0.93108  -1.008   0.3135
interviewee_roledfo              -0.20537    0.95368  -0.215   0.8295
interviewee_roleNGO_rep           0.90332    1.32060   0.684   0.4940
interviewee_gendermale           -0.26549    0.65037  -0.408   0.6831
interviewee_age                  -0.01067    0.02092  -0.510   0.6099
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.536
intrvw_rlt_ -0.402  0.756
```

```
intrvw_rldf -0.591  0.722  0.688
intrvw_NGO_ -0.422  0.513  0.477  0.518
intrvw_gndr -0.358  0.098  0.032 -0.014    0.018
interview_g -0.660  0.015 -0.181  0.194    0.147 -0.090
optimizer (Nelder_Mead) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.0473226 (tol = 0.002, component 1)

 contrast                        odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member      2.123 1.952 Inf    1   0.819  0.9249
 cfmg_member / trad_leader           2.556 2.380 Inf    1   1.008  0.8519
 cfmg_member / dfo                   1.228 1.171 Inf    1   0.215  0.9995
 cfmg_member / NGO_rep               0.405 0.535 Inf    1  -0.684  0.9600
 cfmg_exec_member / trad_leader      1.204 0.779 Inf    1   0.287  0.9985
 cfmg_exec_member / dfo              0.578 0.404 Inf    1  -0.784  0.9354
 cfmg_exec_member / NGO_rep          0.191 0.221 Inf    1  -1.429  0.6087
 trad_leader / dfo                   0.480 0.358 Inf    1  -0.985  0.8622
 trad_leader / NGO_rep               0.159 0.190 Inf    1  -1.535  0.5393
 dfo / NGO_rep                       0.330 0.383 Inf    1  -0.955  0.8752

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

Carbon income benefit sharing

Company-community arrangements

CFMG executive FGDs reported on the components were included in their carbon finance benefit-sharing arrangement with the carbon trading enterprise (Figure 2). Clearly the agreements are dominated by revenue sharing, with relatively few also containing agreements on employment, infrastructure development and livelihood support.

```r
prefix_bs <- "economic_transformation_benefit_sharing_arrangement_company_community_"

bs_labels <- c(
  "revenue_sharing_percentage_of_carbon_finance_proceeds_given_to_the_community"
  "employment_opportunities_for_community_members"
  "investment_in_community_infrastructure_e_g_schools_clinics_roads"
  "support_for_sustainable_livelihoods_e_g_alternative_income_generating_activities_training_
)
```

```r
full_bs_vars <- paste0(prefix_bs, names(bs_labels))

cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(all_of(full_bs_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "n") |>
  mutate(label = factor(bs_labels[sub(prefix_bs, "", variable, fixed = TRUE)],
                        levels = rev(bs_labels))) |>
  ggplot(aes(x = n, y = label)) +
  geom_col(fill = "#2A9D8F", width = 0.55) +
  geom_text(aes(label = n), hjust = -0.3, size = 5, colour = "grey30") +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.15)),
    name   = "Number of CFMGs"
  ) +
  scale_y_discrete(expand = expansion(add = 0.5)) +
  labs(y = NULL) +
  theme_minimal(base_size = 18) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.x        = element_text(colour = "grey20"),
    axis.text.y        = element_text(size = 14, colour = "grey20"),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

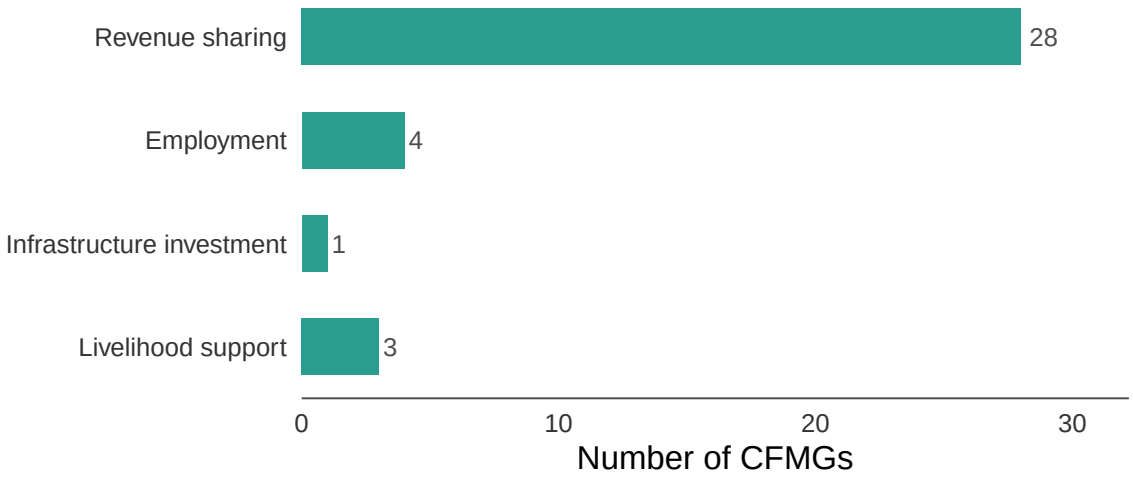

Figure 2: Components included in the carbon finance benefit-sharing arrangement between the carbon company and the community as reported by CFMG executive FGDs. The sample includes only those CFMGs that stated they had an agreement with a carbon trading company.

Within-community sharing

We also asked FGD groups about the carbon income sharing arrangements within the community. We first asked who is responsible for the decision-making (Figure 3) and then we asked on what principles the benefit sharing was based (Figure 4). As can be seen there is broad agreement among the FGD groups on who makes the decision about the caron benefit sharing arrangements. In approximately half the CFs the CFMG executive is responsible for the decision, while in the other half the entire CFMG is responsible.

```
prefix_bsc <- "economic_transformation_benefit_sharing_arrangement_carbon_income_"

decision_labels <- c(
  "decided_by_the_whole_cfmg"     = "Whole CFMG",
  "decided_by_the_cfmg_executive" = "Executive decision"
)

full_decision_vars <- paste0(prefix_bsc, names(decision_labels))

cfmg |>
  filter(interview_type %in% c("cfmg_executive", "cfmg_female",
                               "cfmg_male", "cfmg_youth")) |>
```

```r
mutate(group = factor(
  case_match(
    interview_type,
    "cfmg_executive" ~ "Executive",
    "cfmg_female"    ~ "Female",
    "cfmg_male"      ~ "Male",
    "cfmg_youth"     ~ "Youth"
  ),
  levels = c("Youth", "Male", "Female", "Executive")
)) |>
select(group, all_of(full_decision_vars)) |>
pivot_longer(all_of(full_decision_vars), names_to = "variable", values_to = "val") |>
filter(val == 1) |>
mutate(mechanism = factor(
  decision_labels[sub(prefix_bsc, "", variable, fixed = TRUE)],
  levels = decision_labels
)) |>
count(group, mechanism) |>
ggplot(aes(y = group, x = n, fill = mechanism)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.3) +
geom_text(
  aes(label = n),
  position = position_stack(vjust = 0.5),
  size = 3.5, colour = "white", fontface = "bold"
) +
scale_fill_manual(
  values = c(
    "Whole CFMG"         = "#2A9D8F",
    "Executive decision" = "#E07B54"
  ),
  guide = guide_legend(reverse = TRUE)
) +
scale_x_continuous(
  expand = expansion(mult = c(0, 0.05)),
  name   = "Number of FGDs"
) +
scale_y_discrete(expand = expansion(add = 0.5)) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
```

```
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.x        = element_text(colour = "grey20"),
    axis.text.y        = element_text(size = 14, colour = "grey20"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(0.85)),
    plot.margin        = margin(10, 15, 10, 10)
  )
```

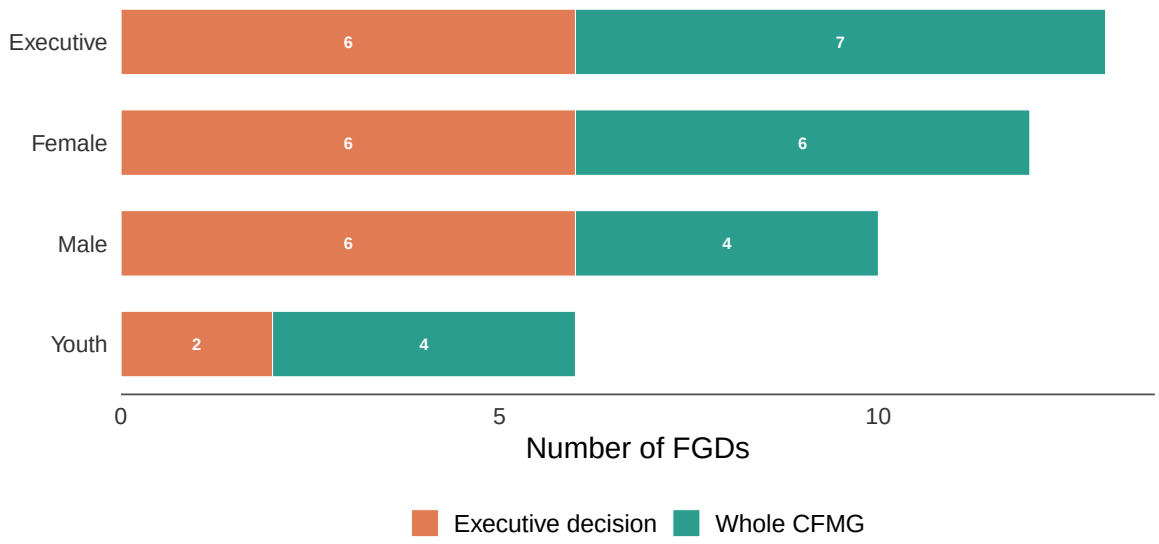

Figure 3: Who decides how carbon income is shared within the community by FGD group. The length of the segments is equal to the number of times that option was mentioned by the FGD group. Note: for 12 CFs the benefit-sharing arrangement is defined in the CFMG constitution.

```
prefix_bsc <- "economic_transformation_benefit_sharing_arrangement_carbon_income_"

principle_labels <- c(
  "equally_among_all_members_regardless_of_gender_or_group"                          = "Equ
  "based_on_contributions_or_roles_in_cfmg_activities"                               = "By
  "prioritized_for_marginalized_groups_e_g_women_youth_aged_persons_with_disabilities" = "Mar
)

full_principle_vars <- paste0(prefix_bsc, names(principle_labels))
```

```r
cfmg |>
  filter(interview_type %in% c("cfmg_executive", "cfmg_female",
                               "cfmg_male", "cfmg_youth")) |>
  mutate(group = factor(
    case_match(
      interview_type,
      "cfmg_executive" ~ "Executive",
      "cfmg_female"    ~ "Female",
      "cfmg_male"      ~ "Male",
      "cfmg_youth"     ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female", "Executive")
  )) |>
  select(group, all_of(full_principle_vars)) |>
  pivot_longer(all_of(full_principle_vars), names_to = "variable", values_to = "val") |>
  filter(val == 1) |>
  mutate(mechanism = factor(
    principle_labels[sub(prefix_bsc, "", variable, fixed = TRUE)],
    levels = principle_labels
  )) |>
  count(group, mechanism) |>
  ggplot(aes(y = group, x = n, fill = mechanism)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.3) +
  geom_text(
    aes(label = n),
    position = position_stack(vjust = 0.5),
    size = 3.5, colour = "white", fontface = "bold"
  ) +
  scale_fill_manual(
    values = c(
      "Equal shares"        = "#2A9D8F",
      "By contribution"     = "#57B8B0",
      "Marginalised groups" = "#E9C46A"
    ),
    guide = guide_legend(reverse = TRUE)
  ) +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.05)),
    name   = "Number of FGDs"
  ) +
  scale_y_discrete(expand = expansion(add = 0.5)) +
  labs(y = NULL, fill = NULL) +
```

```r
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.x        = element_text(colour = "grey20"),
  axis.text.y        = element_text(size = 14, colour = "grey20"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(0.85)),
  plot.margin        = margin(10, 15, 10, 10)
)
```

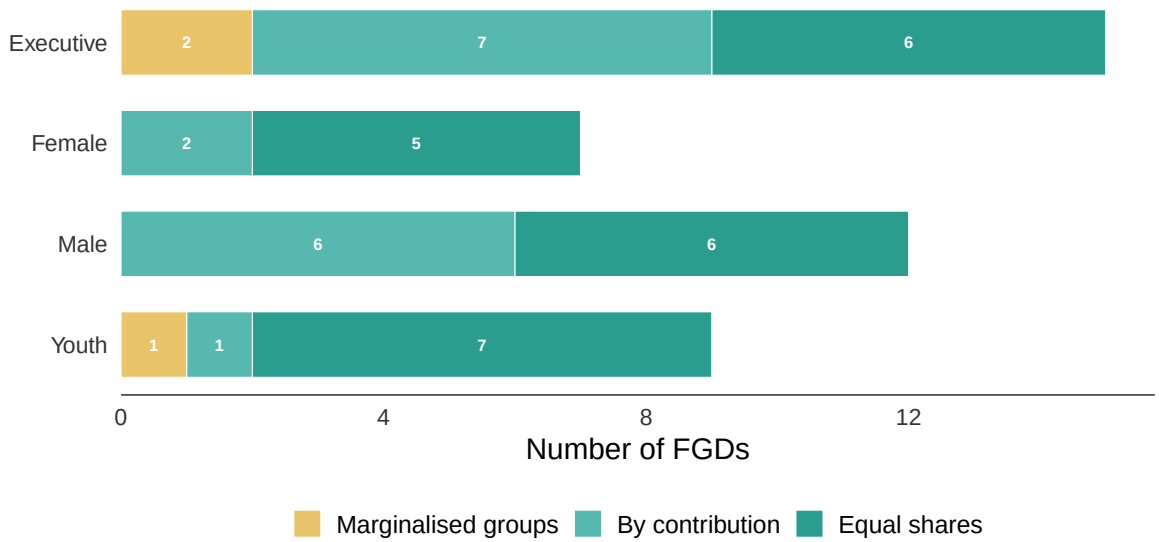

Figure 4: Principles governing carbon income sharing within the community by FGD group. The length of the segments is equal to the number of times that option was mentioned by the FGD group. Note: for 12 CFs the benefit-sharing arrangement is defined in the CFMG constitution.

According to the responses of the CFMG executives, seven CFs shared benefits acccording to members contribution to CFM while six shared them equally among the community households. Only two said that consideration of marginalised or vulnerable groups was a consideration. The other FGD groups were broadly similar in their responses and differences may be due to their knowledge of the arrangements, rather than actual differences in the arrangements.

External determination of benefit sharing

We also asked FGD groups whether stakeholders outside the community were also involved in determining the benefit-sharing arrangement (Figure 5).

```
prefix_bse <- "economic_transformation_benefit_sharing_mechanism_determined_by_"

bse_labels <- c(
  "carbon_trader"                        = "Carbon trader",
  "the_forestry_department_fd"           = "Forestry Dept",
  "an_ngo_or_external_organization"      = "NGO",
  "a_multiple_stakeholder_joint_decision"= "Joint decision",
  "traditional_leaders"                  = "Traditional leaders"
)

full_bse_vars <- paste0(prefix_bse, names(bse_labels))

cfmg |>
  filter(interview_type %in% c("cfmg_executive", "cfmg_female",
                               "cfmg_male", "cfmg_youth")) |>
  mutate(group = factor(
    case_match(
      interview_type,
      "cfmg_executive" ~ "Executive",
      "cfmg_female"    ~ "Female",
      "cfmg_male"      ~ "Male",
      "cfmg_youth"     ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female", "Executive")
  )) |>
  select(group, all_of(full_bse_vars)) |>
  pivot_longer(all_of(full_bse_vars), names_to = "variable", values_to = "val") |>
  filter(val == 1) |>
  mutate(mechanism = factor(
    bse_labels[sub(prefix_bse, "", variable, fixed = TRUE)],
    levels = bse_labels
  )) |>
  count(group, mechanism) |>
  ggplot(aes(y = group, x = n, fill = mechanism)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.3) +
  geom_text(
    aes(label = n),
    position = position_stack(vjust = 0.5),
```

```r
  size = 3.5, colour = "white", fontface = "bold"
) +
scale_fill_manual(
  values = c(
    "Carbon trader"      = "#2A9D8F",
    "Forestry Dept"      = "#57B8B0",
    "NGO"                = "#E9C46A",
    "Joint decision"     = "#F4A261",
    "Traditional leaders"= "#E07B54"
  ),
  guide = guide_legend(reverse = TRUE, nrow = 2, byrow = TRUE)
) +
scale_x_continuous(
  expand = expansion(mult = c(0, 0.05)),
  name   = "Number of FGDs"
) +
scale_y_discrete(expand = expansion(add = 0.5)) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.x        = element_text(colour = "grey20"),
  axis.text.y        = element_text(size = 14, colour = "grey20"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(0.85)),
  plot.margin        = margin(10, 15, 10, 10)
)
```

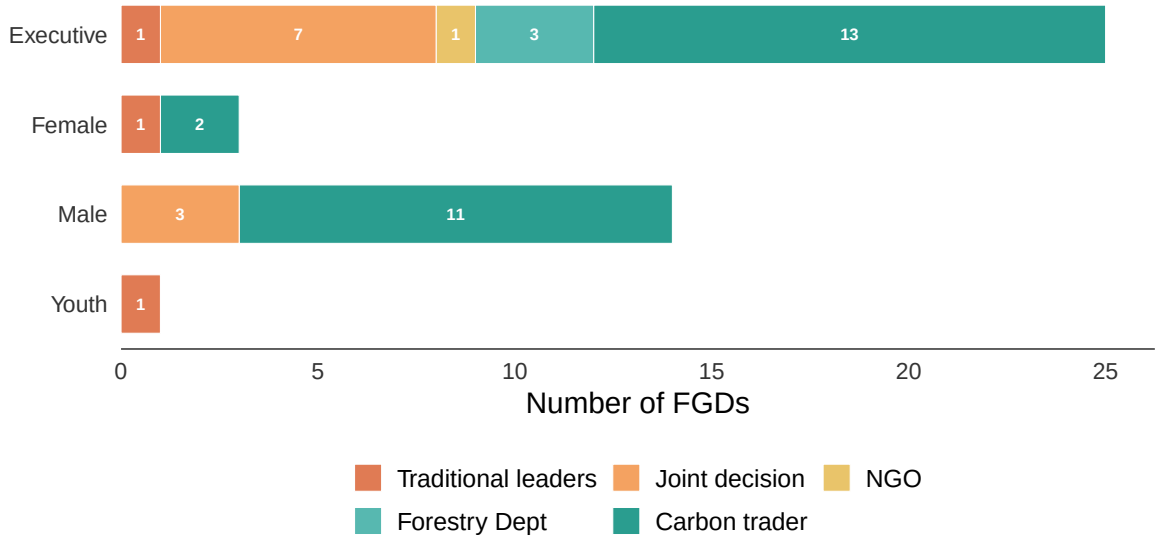

Figure 5: Who determines the benefit-sharing arrangement outside the community, by FGD group.

Gender equity

We asked FGD groups whether women benefit from CFM (Figure 6).

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = equity_and_inclusion_women_benefit_from_cfm,
    response = case_match(response,
                          "yes" ~ "Yes",
                          "no" ~ "No",
                          .default = NA),
    response = factor(response, levels = c("Yes", "No")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
```

```r
    levels = c("Youth", "Male", "Female", "Executive")
  )
) |>
drop_na(response) |>
count(interview_type, response) |>
group_by(interview_type) |>
mutate(prop = n / sum(n)) |>
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
  guide  = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

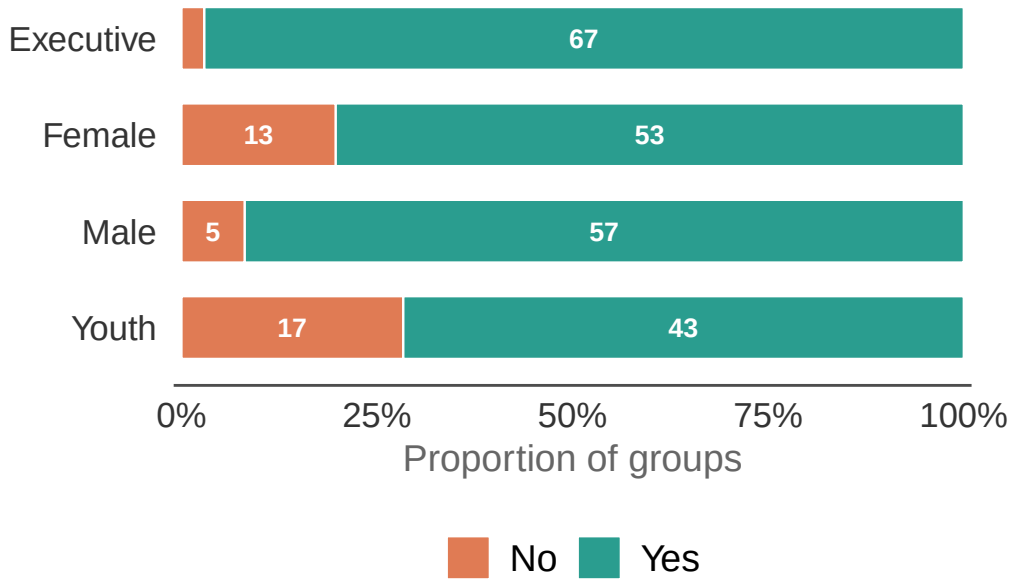

Figure 6: Do women benefit from CFM? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

> **Women benefit from CFM - FGDs - Model methods**
>
> To examine differences among FGD types in whether they agreed that women benefit from CFM, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    empl = equity_and_inclusion_women_benefit_from_cfm,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interview_type = factor(
      case_match(
```

```r
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M3 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M3      <- emmeans(M3, specs = ~ interview_type, type = "response")
pairwise.M3 <- pairs(emm.M3, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   198.8    220.1    -93.4    186.8      251

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.6810  0.1171  0.2150  0.3924  0.7234

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 1.028e+00 1.014e+00
 province      (Intercept) 7.372e-09 8.586e-05
Number of obs: 257, groups:  province_cfmg, 74; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)            3.9491     0.8167   4.835 1.33e-06 ***
interview_typeFemale  -2.2767     0.8216  -2.771 0.005589 **
interview_typeMale    -1.0732     0.8790  -1.221 0.222121
```

```
interview_typeYouth   -2.8156     0.8231  -3.421 0.000625 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.880
intrvw_typM -0.756  0.748
intrvw_typY -0.899  0.841  0.746
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

 contrast           odds.ratio     SE  df null z.ratio p.value
 Executive / Female       9.74  8.006 Inf    1   2.771  0.0286
 Executive / Male         2.92  2.571 Inf    1   1.221  0.6135
 Executive / Youth       16.70 13.748 Inf    1   3.421  0.0035
 Female / Male            0.30  0.182 Inf    1  -1.985  0.1935
 Female / Youth           1.71  0.796 Inf    1   1.160  0.6519
 Male / Youth             5.71  3.479 Inf    1   2.860  0.0220

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether women benefit from CFM. Table 2 shows the percentage reporting this broken down by interviewee role.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = equity_and_inclusion_women_benefit_from_cfm
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
```

```r
    "Community member", "CFMG executive member", "Traditional leader",
    "District Forestry Officer", "NGO representative"
  )),
  empl = case_match(empl,
                    "yes" ~ 1,
                    "no" ~ 0,
                    .default = NA)
) |>
group_by(interviewee_role) |>
summarise(
  n             = n(),
  Women_benefit = mean(empl, na.rm = TRUE),
  .groups       = "drop"
) |>
arrange(interviewee_role) |>
gt() |>
cols_label(
  interviewee_role = "Role",
  n                = "n",
  Women_benefit    = "Agree women benefit from CFM"
) |>
fmt_percent(
  columns  = Women_benefit,
  decimals = 0
) |>
tab_header(
  title    = "Women benefit from CFM, as reported by individual interviewees",
  subtitle = "Values represent the % of positive responses. n = the total number of interv
) |>
opt_align_table_header(align = "left") |>
cols_align(align = "left",   columns = interviewee_role) |>
cols_align(align = "center", columns = c(n, Women_benefit)) |>
tab_style(
  style     = cell_text(align = "center"),
  locations = cells_column_labels(columns = interviewee_role)
) |>
sub_missing()
```

Table 2: Percentage of individual interviewees reporting that women benefit from CFM by interviewee role.

Women benefit from CFM, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses. A positive response indicates that women benefit from CFM.

Role	n	Agree women benefit from CFM
Community member	31	84%
CFMG executive member	63	95%
Traditional leader	75	79%
District Forestry Officer	56	94%
NGO representative	32	100%

Women benefit from CFM - Individual interviewees - Model methods

To examine differences among individual interviewee roles in whether they agreed that women benefit from CFM, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    empl               = equity_and_inclusion_women_benefit_from_cfm,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M4 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
              (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M4      <- emmeans(M4, specs = ~ interviewee_role, type = "response")
pairwise.M4 <- pairs(emm.M4, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
```

```
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   150.7    182.0    -66.3    132.7      230

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.3204  0.0500  0.0840  0.1737  1.4644

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 7.489e+00 2.737e+00
 province      (Intercept) 4.004e-09 6.328e-05
Number of obs: 239, groups:  province_cfmg, 73; province, 8

Fixed effects:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                        1.82010    2.17983   0.835    0.404
interviewee_rolecfmg_exec_member   0.84923    1.15271   0.737    0.461
interviewee_roletrad_leader       -1.24851    1.08807  -1.147    0.251
interviewee_roledfo                1.66136    1.14094   1.456    0.145
interviewee_roleNGO_rep           24.17559  457.94680   0.053    0.958
interviewee_gendermale            -0.51953    0.94133  -0.552    0.581
interviewee_age                    0.04801    0.03221   1.491    0.136

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.501
intrvw_rlt_ -0.385  0.573
intrvw_rldf -0.243  0.383  0.312
intrvw_NGO_  0.000  0.000  0.000  0.000
intrvw_gndr -0.499  0.228  0.074 -0.017    0.000
interview_g -0.214 -0.108 -0.478  0.292    0.000 -0.072
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

 contrast                       odds.ratio     SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member     0.4277 0.4931 Inf    1  -0.737  0.9479
```

```
 cfmg_member / trad_leader          3.4851 3.7921 Inf    1   1.147  0.7812
 cfmg_member / dfo                  0.1899 0.2166 Inf    1  -1.456  0.5911
 cfmg_member / NGO_rep              0.0000 0.0000 Inf    1  -0.053  1.0000
 cfmg_exec_member / trad_leader     8.1477 8.4465 Inf    1   2.024  0.2546
 cfmg_exec_member / dfo             0.4439 0.5657 Inf    1  -0.637  0.9690
 cfmg_exec_member / NGO_rep         0.0000 0.0000 Inf    1  -0.051  1.0000
 trad_leader / dfo                  0.0545 0.0713 Inf    1  -2.224  0.1707
 trad_leader / NGO_rep              0.0000 0.0000 Inf    1  -0.056  1.0000
 dfo / NGO_rep                      0.0000 0.0000 Inf    1  -0.049  1.0000

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

Women disproportionately suffer opportunity costs

We asked FGD groups whether women disproportionately suffer the opportunity costs of CFM (Figure 7).

```r
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = equity_and_inclusion_women_disproportionally_suffer_opportunity_costs,
    response = case_match(response,
                          "yes" ~ "Yes",
                          "no" ~ "No",
                          .default = NA),
    response = factor(response, levels = c("Yes", "No")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
  ) |>
  drop_na(response) |>
  count(interview_type, response) |>
  group_by(interview_type) |>
```

```r
  mutate(prop = n / sum(n)) |>
  ungroup() |>
  ggplot(aes(y = interview_type, x = prop, fill = response)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_text(
    aes(label = ifelse(prop >= 0.05, n, "")),
    position = position_stack(vjust = 0.5),
    size = 3.5, fontface = "bold", colour = "white"
  ) +
  scale_x_continuous(
    labels = percent_format(accuracy = 1),
    expand = expansion(mult = 0.01),
    name   = "Proportion of groups"
  ) +
  scale_fill_manual(
    values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
    guide  = guide_legend(reverse = TRUE)
  ) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

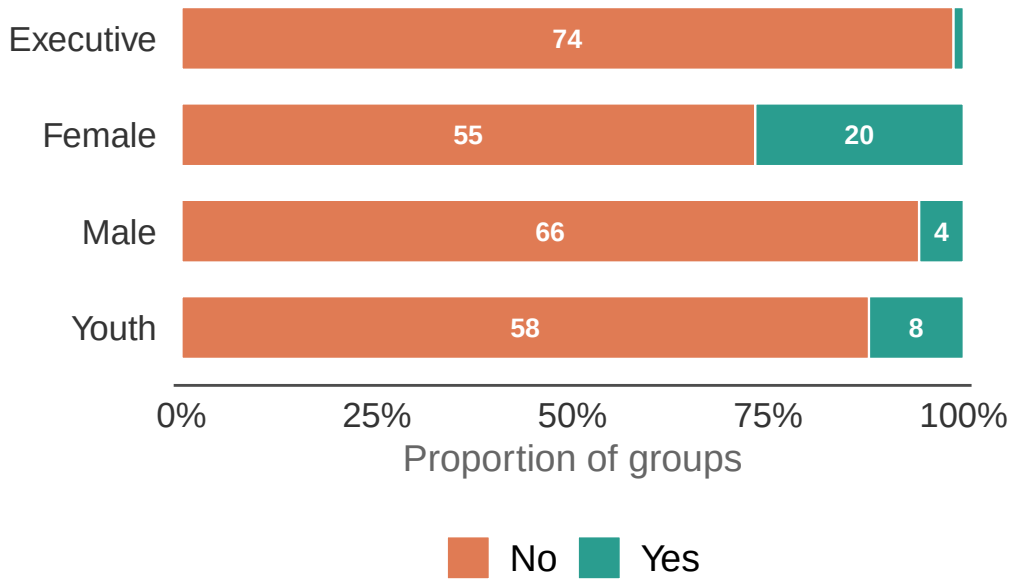

Figure 7: Do women disproportionately suffer the opportunity costs of CFM? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Women opportunity costs - FGDs - Model methods

To examine differences among FGD types in whether they agreed that women disproportionately suffer the opportunity costs of CFM, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    empl = equity_and_inclusion_women_disproportionally_suffer_opportunity_costs,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interview_type = factor(
      case_match(
```

```r
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M5 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M5      <- emmeans(M5, specs = ~ interview_type, type = "response")
pairwise.M5 <- pairs(emm.M5, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   187.2    209.2    -87.6    175.2      280

Scaled residuals:
    Min      1Q  Median      3Q     Max
-0.7762 -0.3908 -0.2364 -0.1061  9.9790

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 2.224e-09 4.716e-05
 province      (Intercept) 2.823e-01 5.313e-01
Number of obs: 286, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)            -4.400      1.023  -4.302 1.69e-05 ***
interview_typeFemale    3.324      1.034   3.213  0.00131 **
interview_typeMale      1.443      1.124   1.284  0.19909
```

```
interview_typeYouth     2.323      1.069   2.173  0.02975 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.947
intrvw_typM -0.865  0.857
intrvw_typY -0.913  0.903  0.830
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

 contrast           odds.ratio     SE  df null z.ratio p.value
 Executive / Female      0.036 0.0373 Inf    1  -3.213  0.0072
 Executive / Male        0.236 0.2654 Inf    1  -1.284  0.5731
 Executive / Youth       0.098 0.1047 Inf    1  -2.173  0.1307
 Female / Male           6.556 3.8319 Inf    1   3.217  0.0071
 Female / Youth          2.721 1.2664 Inf    1   2.151  0.1372
 Male / Youth            0.415 0.2662 Inf    1  -1.371  0.5176

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether women disproportionately suffer the opportunity costs of CFM. Table 3 shows the percentage reporting this broken down by interviewee role.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = equity_and_inclusion_women_disproportionally_suffer_opportunity_costs
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
```

```r
      "Community member", "CFMG executive member", "Traditional leader",
      "District Forestry Officer", "NGO representative"
    )),
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA)
  ) |>
  group_by(interviewee_role) |>
  summarise(
    n         = n(),
    Opp_costs = mean(empl, na.rm = TRUE),
    .groups   = "drop"
  ) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(
    interviewee_role = "Role",
    n                = "n",
    Opp_costs        = "Agree women disproportionately suffer opportunity costs"
  ) |>
  fmt_percent(
    columns  = Opp_costs,
    decimals = 0
  ) |>
  tab_header(
    title    = "Women disproportionately suffer opportunity costs, as reported by individual
    subtitle = "Values represent the % of positive responses. n = the total number of interv
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "left",   columns = interviewee_role) |>
  cols_align(align = "center", columns = c(n, Opp_costs)) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 3: Percentage of individual interviewees reporting that women disproportionately suffer the opportunity costs of CFM by interviewee role.

Women disproportionately suffer opportunity costs, as reported by individual interviewees
Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Agree women disproportionately suffer opportunity costs
Community member	31	29%
CFMG executive member	63	18%
Traditional leader	75	18%
District Forestry Officer	56	21%
NGO representative	32	7%

Women opportunity costs - Individual interviewees - Model methods

To examine differences among individual interviewee roles in whether they agreed that women disproportionately suffer the opportunity costs of CFM, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    empl               = equity_and_inclusion_women_disproportionally_suffer_opportunity_cos
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M6 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
              (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M6      <- emmeans(M6, specs = ~ interviewee_role, type = "response")
pairwise.M6 <- pairs(emm.M6, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
```

```
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   233.9    265.2   -107.9    215.9      230

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.0511 -0.4697 -0.3927 -0.2363  4.9262

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 1.140e-07 0.0003376
 province      (Intercept) 2.931e-01 0.5414223
Number of obs: 239, groups:  province_cfmg, 73; province, 8

Fixed effects:
                                 Estimate Std. Error z value Pr(>|z|)
(Intercept)                      -2.04435    0.97648  -2.094   0.0363 *
interviewee_rolecfmg_exec_member -0.79239    0.58628  -1.352   0.1765
interviewee_roletrad_leader      -0.94637    0.58448  -1.619   0.1054
interviewee_roledfo              -0.21708    0.60936  -0.356   0.7217
interviewee_roleNGO_rep          -1.45769    0.88421  -1.649   0.0992 .
interviewee_gendermale           -0.54889    0.43056  -1.275   0.2024
interviewee_age                   0.03139    0.01529   2.053   0.0401 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.341
intrvw_rlt_ -0.162  0.615
intrvw_rldf -0.545  0.596  0.550
intrvw_NGO_ -0.387  0.385  0.366  0.420
intrvw_gndr -0.286 -0.030 -0.082 -0.045    0.062
interview_g -0.803 -0.007 -0.215  0.257    0.155 -0.035

 contrast                       odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member      2.209 1.295 Inf    1   1.352  0.6588
```

```
 cfmg_member / trad_leader          2.576 1.506 Inf    1   1.619  0.4850
 cfmg_member / dfo                  1.242 0.757 Inf    1   0.356  0.9966
 cfmg_member / NGO_rep              4.296 3.799 Inf    1   1.649  0.4662
 cfmg_exec_member / trad_leader     1.166 0.599 Inf    1   0.300  0.9982
 cfmg_exec_member / dfo             0.563 0.302 Inf    1  -1.070  0.8221
 cfmg_exec_member / NGO_rep         1.945 1.657 Inf    1   0.781  0.9362
 trad_leader / dfo                  0.482 0.273 Inf    1  -1.287  0.6990
 trad_leader / NGO_rep              1.667 1.439 Inf    1   0.592  0.9762
 dfo / NGO_rep                      3.458 2.893 Inf    1   1.483  0.5739

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

Women participate in decision making

We asked FGD groups whether women participate in CFM decision making (Figure 8).

```r
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = equity_and_inclusion_women_participate_in_decision_making,
    response = case_match(response,
                          "yes" ~ "Yes",
                          "no" ~ "No",
                          .default = NA),
    response = factor(response, levels = c("Yes", "No")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
  ) |>
  drop_na(response) |>
  count(interview_type, response) |>
  group_by(interview_type) |>
  mutate(prop = n / sum(n)) |>
```

```r
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
  guide  = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

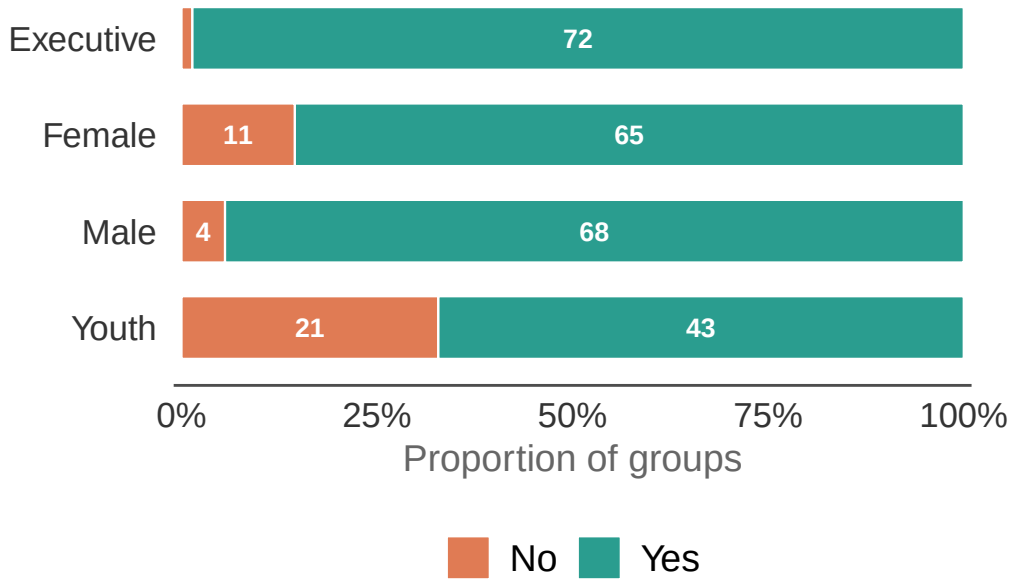

Figure 8: Do women participate in CFM decision making? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

> **Women decision making - FGDs - Model methods**
>
> To examine differences among FGD types in whether they agreed that women participate in CFM decision making, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    empl = equity_and_inclusion_women_participate_in_decision_making,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interview_type = factor(
      case_match(
```

```r
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M7 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M7      <- emmeans(M7, specs = ~ interview_type, type = "response")
pairwise.M7 <- pairs(emm.M7, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   185.5    207.4    -86.7    173.5      279

Scaled residuals:
    Min      1Q  Median      3Q     Max
-7.2946  0.0754  0.1654  0.3442  1.0360

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 1.2892   1.135
 province      (Intercept) 0.8483   0.921
Number of obs: 285, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)             5.022      1.180   4.257 2.07e-05 ***
interview_typeFemale   -2.534      1.091  -2.322 0.020258 *
interview_typeMale     -1.238      1.167  -1.060 0.288959
```

```
interview_typeYouth    -4.045      1.116  -3.625 0.000289 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.839
intrvw_typM -0.746  0.826
intrvw_typY -0.903  0.877  0.793

 contrast           odds.ratio     SE  df null z.ratio p.value
 Executive / Female     12.598 13.749 Inf    1   2.322  0.0931
 Executive / Male        3.447  4.023 Inf    1   1.060  0.7137
 Executive / Youth      57.108 63.717 Inf    1   3.625  0.0016
 Female / Male           0.274  0.183 Inf    1  -1.935  0.2133
 Female / Youth          4.533  2.485 Inf    1   2.757  0.0298
 Male / Youth           16.566 12.195 Inf    1   3.813  0.0008

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether women participate in CFM decision making. Table 4 shows the percentage reporting this broken down by interviewee role.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = equity_and_inclusion_women_participate_in_decision_making
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
      "Community member", "CFMG executive member", "Traditional leader",
      "District Forestry Officer", "NGO representative"
```

```r
    )),
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA)
  ) |>
  group_by(interviewee_role) |>
  summarise(
    n               = n(),
    Decision_making = mean(empl, na.rm = TRUE),
    .groups         = "drop"
  ) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(
    interviewee_role = "Role",
    n                = "n",
    Decision_making  = "Agree women participate in decision making"
  ) |>
  fmt_percent(
    columns  = Decision_making,
    decimals = 0
  ) |>
  tab_header(
    title    = "Women participate in CFM decision making, as reported by individual intervie
    subtitle = "Values represent the % of positive responses. n = the total number of interv
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "left",   columns = interviewee_role) |>
  cols_align(align = "center", columns = c(n, Decision_making)) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 4: Percentage of individual interviewees reporting that women participate in CFM decision making by interviewee role.

Women participate in CFM decision making, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Agree women participate in decision making
Community member	31	100%
CFMG executive member	63	95%
Traditional leader	75	92%
District Forestry Officer	56	92%
NGO representative	32	97%

Women decision making - Individual interviewees - Model methods

To examine differences among individual interviewee roles in whether they agreed that women participate in CFM decision making, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    empl               = equity_and_inclusion_women_participate_in_decision_making,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M8 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
              (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M8      <- emmeans(M8, specs = ~ interviewee_role, type = "response")
pairwise.M8 <- pairs(emm.M8, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
```

```
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   117.9    149.7    -49.9     99.9      243

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.6939  0.1089  0.1823  0.2498  0.5835

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.3590   0.5992
 province      (Intercept) 0.8907   0.9438
Number of obs: 252, groups:  province_cfmg, 74; province, 8

Fixed effects:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                      2.162e+01  1.407e+04   0.002    0.999
interviewee_rolecfmg_exec_member -1.916e+01  1.407e+04  -0.001    0.999
interviewee_roletrad_leader      -2.023e+01  1.407e+04  -0.001    0.999
interviewee_roledfo              -1.952e+01  1.407e+04  -0.001    0.999
interviewee_roleNGO_rep          -1.883e+01  1.407e+04  -0.001    0.999
interviewee_gendermale            2.461e-01  7.490e-01   0.329    0.742
interviewee_age                   1.869e-02  2.976e-02   0.628    0.530

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -1.000
intrvw_rlt_ -1.000  1.000
intrvw_rldf -1.000  1.000  1.000
intrvw_NGO_ -1.000  1.000  1.000  1.000
intrvw_gndr  0.000  0.000  0.000  0.000    0.000
interview_g  0.000  0.000  0.000  0.000    0.000  0.058
optimizer (Nelder_Mead) convergence code: 0 (OK)
unable to evaluate scaled gradient
Model failed to converge: degenerate  Hessian with 1 negative eigenvalues

 contrast                       odds.ratio       SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member   2.09e+08 2.94e+12 Inf    1   0.001  1.0000
 cfmg_member / trad_leader        6.08e+08 8.56e+12 Inf    1   0.001  1.0000
```

```
 cfmg_member / dfo                3.01e+08 4.24e+12 Inf    1   0.001  1.0000
 cfmg_member / NGO_rep            1.51e+08 2.12e+12 Inf    1   0.001  1.0000
 cfmg_exec_member / trad_leader   3.00e+00 3.00e+00 Inf    1   1.172  0.7672
 cfmg_exec_member / dfo           1.00e+00 1.00e+00 Inf    1   0.408  0.9942
 cfmg_exec_member / NGO_rep       1.00e+00 1.00e+00 Inf    1  -0.247  0.9992
 trad_leader / dfo                0.00e+00 0.00e+00 Inf    1  -0.769  0.9395
 trad_leader / NGO_rep            0.00e+00 0.00e+00 Inf    1  -1.041  0.8365
 dfo / NGO_rep                    1.00e+00 1.00e+00 Inf    1  -0.553  0.9816

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

Transparency in benefit sharing

We asked FGD groups whether there is transparency in CFM benefit sharing (Figure 9).

```r
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = equity_and_inclusion_transparency_in_benefit_sharing,
    response = case_match(response,
                          "yes" ~ "Yes",
                          "no" ~ "No",
                          .default = NA),
    response = factor(response, levels = c("Yes", "No")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
  ) |>
  drop_na(response) |>
  count(interview_type, response) |>
  group_by(interview_type) |>
  mutate(prop = n / sum(n)) |>
  ungroup() |>
```

```r
ggplot(aes(y = interview_type, x = prop, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
  guide  = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

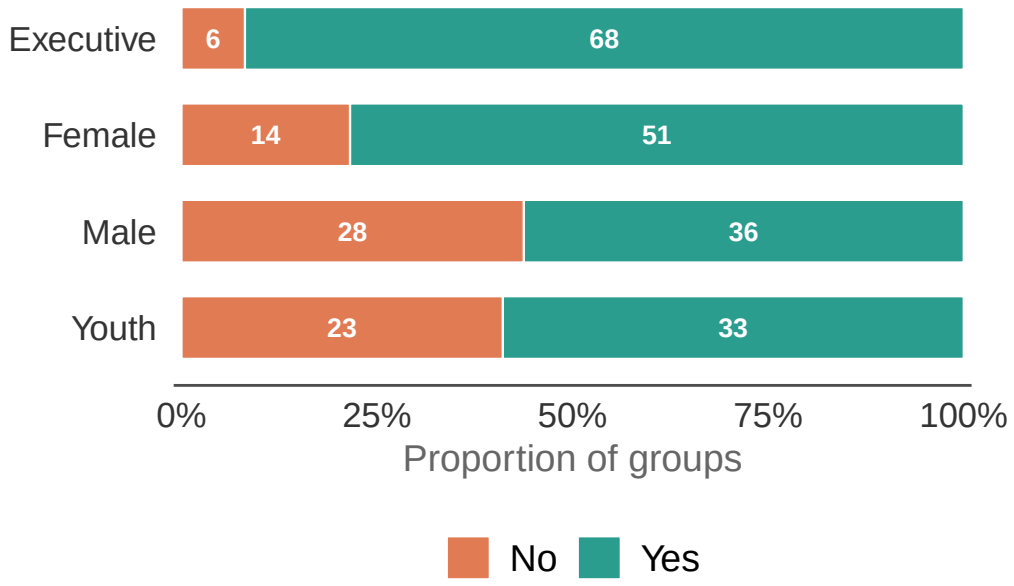

Figure 9: Is there transparency in CFM benefit sharing? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

> **Transparency in benefit sharing - FGDs - Model methods**
>
> To examine differences among FGD types in whether they agreed there is transparency in CFM benefit sharing, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    empl = equity_and_inclusion_transparency_in_benefit_sharing,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interview_type = factor(
      case_match(
```

```r
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M9 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M9      <- emmeans(M9, specs = ~ interview_type, type = "response")
pairwise.M9 <- pairs(emm.M9, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   280.4    301.7   -134.2    268.4      253

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.6334 -0.8479  0.3471  0.6085  1.0986

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.3862   0.6215
 province      (Intercept) 0.2366   0.4865
Number of obs: 259, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)            2.5724     0.5011   5.134 2.84e-07 ***
interview_typeFemale  -1.1834     0.5423  -2.182   0.0291 *
interview_typeMale    -2.2696     0.5312  -4.273 1.93e-05 ***
```

```
interview_typeYouth   -2.1616     0.5401  -4.002 6.28e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.726
intrvw_typM -0.793  0.681
intrvw_typY -0.772  0.667  0.717

 contrast           odds.ratio    SE  df null z.ratio p.value
 Executive / Female      3.265 1.771 Inf    1   2.182  0.1281
 Executive / Male        9.676 5.139 Inf    1   4.273  0.0001
 Executive / Youth       8.685 4.691 Inf    1   4.002  0.0004
 Female / Male           2.963 1.270 Inf    1   2.534  0.0548
 Female / Youth          2.660 1.175 Inf    1   2.215  0.1191
 Male / Youth            0.898 0.362 Inf    1  -0.268  0.9933

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether there is transparency in CFM benefit sharing. Table 5 shows the percentage reporting this broken down by interviewee role.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = equity_and_inclusion_transparency_in_benefit_sharing
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
      "Community member", "CFMG executive member", "Traditional leader",
      "District Forestry Officer", "NGO representative"
```

```
    )),
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA)
  ) |>
  group_by(interviewee_role) |>
  summarise(
    n            = n(),
    Transparency = mean(empl, na.rm = TRUE),
    .groups      = "drop"
  ) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(
    interviewee_role = "Role",
    n                = "n",
    Transparency     = "Agree there is transparency in benefit sharing"
  ) |>
  fmt_percent(
    columns  = Transparency,
    decimals = 0
  ) |>
  tab_header(
    title    = "Transparency in CFM benefit sharing, as reported by individual interviewees"
    subtitle = "Values represent the % of positive responses. n = the total number of interv
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "left",   columns = interviewee_role) |>
  cols_align(align = "center", columns = c(n, Transparency)) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 5: Percentage of individual interviewees reporting that there is transparency in CFM benefit sharing by interviewee role.

Transparency in CFM benefit sharing, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Agree there is transparency in benefit sharing
Community member	31	92%
CFMG executive member	63	93%
Traditional leader	75	76%
District Forestry Officer	56	79%
NGO representative	32	80%

Transparency in benefit sharing - Individual interviewees - Model methods

To examine differences among individual interviewee roles in whether they agreed there is transparency in CFM benefit sharing, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    empl               = equity_and_inclusion_transparency_in_benefit_sharing,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M10 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
               (1 | province) + (1 | province_cfmg),
             data = test.data, family = binomial)

emm.M10      <- emmeans(M10, specs = ~ interviewee_role, type = "response")
pairwise.M10 <- pairs(emm.M10, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
```

```
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   217.3    248.5    -99.7    199.3      228

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.7182  0.1959  0.3126  0.4555  0.8306

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.4647   0.6817
 province      (Intercept) 0.4395   0.6630
Number of obs: 237, groups:  province_cfmg, 74; province, 8

Fixed effects:
                                 Estimate Std. Error z value Pr(>|z|)
(Intercept)                       3.30488    1.41617   2.334   0.0196 *
interviewee_rolecfmg_exec_member  0.07000    0.98122   0.071   0.9431
interviewee_roletrad_leader      -1.52883    0.86864  -1.760   0.0784 .
interviewee_roledfo              -1.38955    0.94083  -1.477   0.1397
interviewee_roleNGO_rep          -1.50598    0.98933  -1.522   0.1280
interviewee_gendermale            0.44467    0.52397   0.849   0.3961
interviewee_age                  -0.01584    0.01852  -0.855   0.3924
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.542
intrvw_rlt_ -0.495  0.729
intrvw_rldf -0.728  0.735  0.783
intrvw_NGO_ -0.691  0.683  0.742  0.782
intrvw_gndr -0.238 -0.046 -0.127 -0.042   -0.040
interview_g -0.760  0.132  0.000  0.333    0.318  0.013

 contrast                        odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member       0.932 0.915 Inf    1  -0.071  1.0000
 cfmg_member / trad_leader            4.613 4.007 Inf    1   1.760  0.3972
 cfmg_member / dfo                    4.013 3.776 Inf    1   1.477  0.5775
 cfmg_member / NGO_rep                4.509 4.460 Inf    1   1.522  0.5479
```

```
 cfmg_exec_member / trad_leader      4.947 3.406 Inf    1   2.322  0.1378
 cfmg_exec_member / dfo              4.304 3.014 Inf    1   2.084  0.2268
 cfmg_exec_member / NGO_rep          4.835 3.792 Inf    1   2.010  0.2611
 trad_leader / dfo                   0.870 0.522 Inf    1  -0.232  0.9994
 trad_leader / NGO_rep               0.977 0.662 Inf    1  -0.034  1.0000
 dfo / NGO_rep                       1.123 0.717 Inf    1   0.182  0.9998

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

We also examined which mechanisms are being used to achieve transparency in benefit sharing among the community female, male, and youth FGDs (Figure 10).

```r
prefix_trans <- "equity_and_inclusion_mechanisms_for_transparency_fairness_"

trans_labels <- c(
  "community_meetings_to_discuss_benefit_sharing_decisions"
  "written_agreements_or_guidelines_outlining_benefit_distribution_rules"
  "memos"
  "financial_reports_or_audits_shared_with_community_members"
  "inclusive_representation_of_women_youth_and_marginalized_groups_in_decision_making"
  "external_oversight_by_government_agencies_ngos_or_independent_bodies"
  "grievance_and_dispute_resolution_mechanisms_for_complaints_or_conflicts"
  "needs_assessment"
)

full_trans_vars <- paste0(prefix_trans, names(trans_labels))

cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  mutate(group = factor(
    case_match(
      interview_type,
      "cfmg_female" ~ "Female",
      "cfmg_male"   ~ "Male",
      "cfmg_youth"  ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female")
  )) |>
  select(group, all_of(full_trans_vars)) |>
  pivot_longer(all_of(full_trans_vars), names_to = "variable", values_to = "val") |>
```

```r
filter(val == 1) |>
mutate(mechanism = factor(
  trans_labels[sub(prefix_trans, "", variable, fixed = TRUE)],
  levels = rev(trans_labels)
)) |>
count(group, mechanism) |>
complete(group, mechanism, fill = list(n = 0)) |>
ggplot(aes(y = mechanism, x = n, fill = group)) +
geom_col(position = position_dodge(width = 0.75), width = 0.65, colour = "white") +
geom_text(
  aes(label = ifelse(n > 0, n, "")),
  position = position_dodge(width = 0.75),
  hjust = -0.3, size = 3.5, colour = "grey30"
) +
scale_fill_manual(
  values = c(
    "Female" = "#2A9D8F",
    "Male"   = "#57B8B0",
    "Youth"  = "#E9C46A"
  ),
  guide = guide_legend(reverse = TRUE)
) +
scale_x_continuous(
  expand = expansion(mult = c(0, 0.15)),
  name   = "Number of FGDs"
) +
scale_y_discrete(labels = \(x) str_wrap(x, width = 25), expand = expansion(add = 0.6)) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.x        = element_text(colour = "grey20"),
  axis.text.y        = element_text(size = 14, colour = "grey20"),
  legend.position    = "bottom",
  plot.margin        = margin(10, 25, 10, 10)
)
```

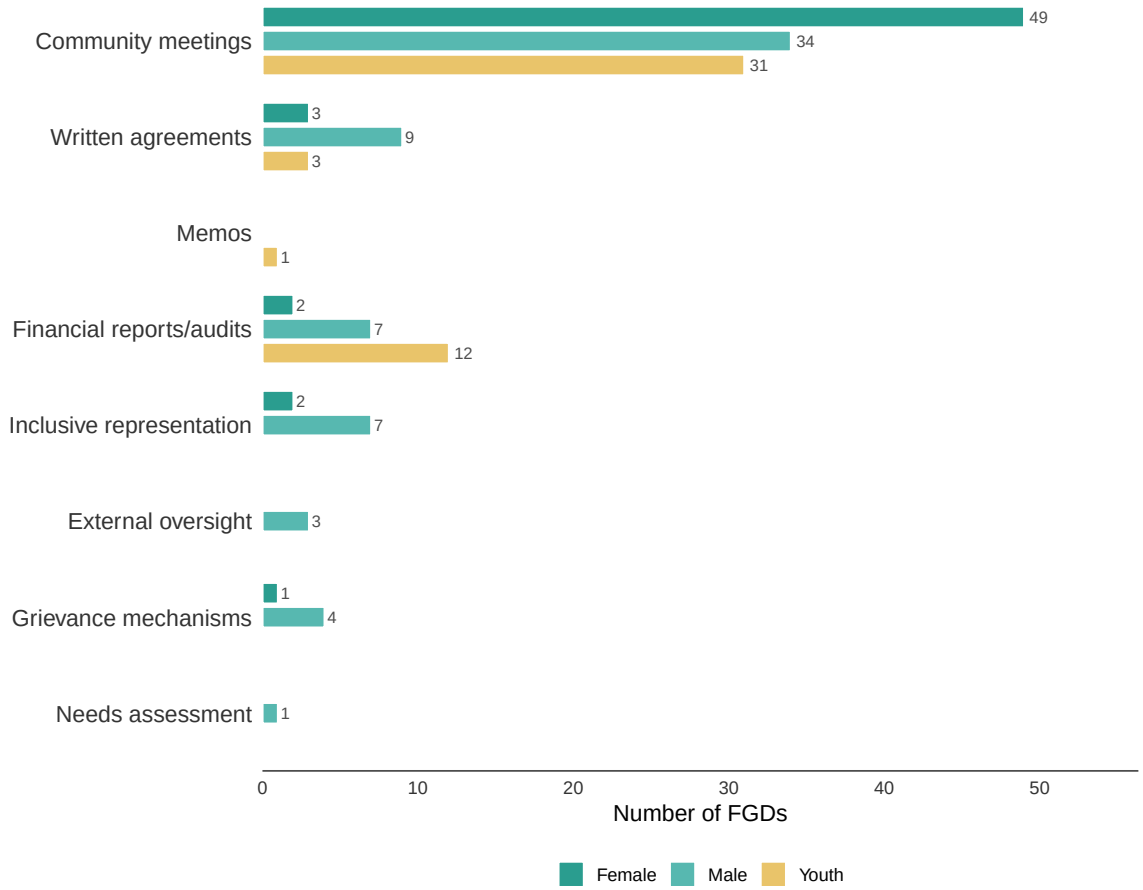

Figure 10: Mechanisms used to achieve transparency in CFM benefit sharing, as reported by community female, male, and youth FGDs. Bars show the number of FGDs reporting each mechanism.

Grievance and Redress

We asked FGD groups whether their CFMG has a grievance and redress mechanism (Figure 11).

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = equity_and_inclusion_grievance_redress_mechanism,
    response = case_match(response,
                          "yes" ~ "Yes",
```

```r
                          "no" ~ "No",
                          .default = NA),
  response = factor(response, levels = c("Yes", "No")),
  interview_type = factor(
    case_match(
      interview_type,
      "cfmg_executive" ~ "Executive",
      "cfmg_female"    ~ "Female",
      "cfmg_male"      ~ "Male",
      "cfmg_youth"     ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female", "Executive")
  )
) |>
drop_na(response) |>
count(interview_type, response) |>
group_by(interview_type) |>
mutate(prop = n / sum(n)) |>
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
  guide  = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
```

```
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

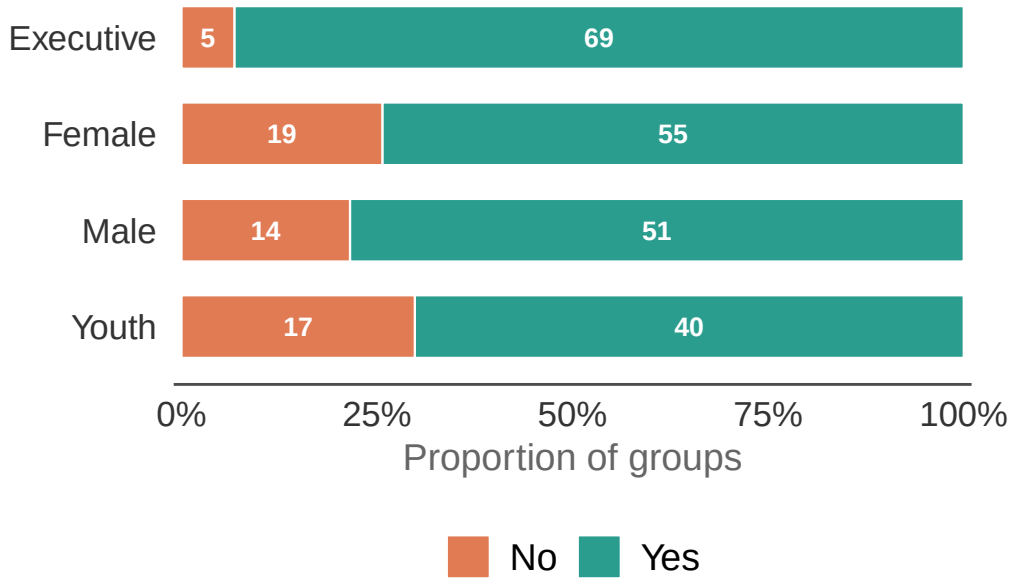

Figure 11: Does the CFMG have a grievance and redress mechanism? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Grievance and redress - FGDs - Model methods

To examine differences among FGD types in whether they reported having a grievance and redress mechanism, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```r
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    empl = equity_and_inclusion_grievance_redress_mechanism,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M11 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
             data = test.data, family = binomial)

emm.M11      <- emmeans(M11, specs = ~ interview_type, type = "response")
pairwise.M11 <- pairs(emm.M11, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   268.3    289.9   -128.1    256.3      264

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.1004  0.2075  0.4423  0.5798  0.7765

Random effects:
```

```
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.05611  0.2369
 province      (Intercept) 0.16831  0.4103
Number of obs: 270, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)            2.6416     0.5132   5.147 2.64e-07 ***
interview_typeFemale  -1.5702     0.5398  -2.909  0.00363 **
interview_typeMale    -1.3029     0.5598  -2.327  0.01994 *
interview_typeYouth   -1.7910     0.5551  -3.226  0.00125 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.809
intrvw_typM -0.784  0.723
intrvw_typY -0.793  0.730  0.704


 contrast           odds.ratio    SE  df null z.ratio p.value
 Executive / Female      4.807 2.595 Inf    1   2.909  0.0190
 Executive / Male        3.680 2.060 Inf    1   2.327  0.0918
 Executive / Youth       5.995 3.328 Inf    1   3.226  0.0069
 Female / Male           0.765 0.314 Inf    1  -0.652  0.9148
 Female / Youth          1.247 0.502 Inf    1   0.549  0.9470
 Male / Youth            1.629 0.699 Inf    1   1.137  0.6666

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether their CFMG has a grievance and redress mechanism. Table 6 shows the percentage reporting this broken down by interviewee role.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = equity_and_inclusion_grievance_redress_mechanism
  ) |>
  mutate(
    interviewee_role = case_match(
```

```
    as.character(interviewee_role),
    "cfmg_member"      ~ "Community member",
    "cfmg_exec_member" ~ "CFMG executive member",
    "trad_leader"      ~ "Traditional leader",
    "dfo"              ~ "District Forestry Officer",
    "NGO_rep"          ~ "NGO representative"
  ),
  interviewee_role = factor(interviewee_role, levels = c(
    "Community member", "CFMG executive member", "Traditional leader",
    "District Forestry Officer", "NGO representative"
  )),
  empl = case_match(empl,
                    "yes" ~ 1,
                    "no" ~ 0,
                    .default = NA)
) |>
group_by(interviewee_role) |>
summarise(
  n         = n(),
  Grievance = mean(empl, na.rm = TRUE),
  .groups   = "drop"
) |>
arrange(interviewee_role) |>
gt() |>
cols_label(
  interviewee_role = "Role",
  n                = "n",
  Grievance        = "Grievance and redress mechanism"
) |>
fmt_percent(
  columns  = Grievance,
  decimals = 0
) |>
tab_header(
  title    = "Grievance and redress mechanism, as reported by individual interviewees",
  subtitle = "Values represent the % of positive responses. n = the total number of interv
) |>
opt_align_table_header(align = "left") |>
cols_align(align = "left",   columns = interviewee_role) |>
cols_align(align = "center", columns = c(n, Grievance)) |>
tab_style(
  style     = cell_text(align = "center"),
```

```
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 6: Percentage of individual interviewees reporting that the CFMG has a grievance and redress mechanism by interviewee role.

Grievance and redress mechanism, as reported by individual interviewees
Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Grievance and redress mechanism
Community member	31	97%
CFMG executive member	63	90%
Traditional leader	75	86%
District Forestry Officer	56	77%
NGO representative	32	86%

Grievance and redress - Individual interviewees - Model methods

To examine differences among individual interviewee roles in whether they reported the CFMG has a grievance and redress mechanism, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    empl               = equity_and_inclusion_grievance_redress_mechanism,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))
```

```r
M12 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
               (1 | province) + (1 | province_cfmg),
             data = test.data, family = binomial)

emm.M12      <- emmeans(M12, specs = ~ interviewee_role, type = "response")
pairwise.M12 <- pairs(emm.M12, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   199.2    230.5    -90.6    181.2      231

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.0701  0.2416  0.3224  0.4368  0.7224

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 1.417e-07 0.0003764
 province      (Intercept) 2.551e-01 0.5050848
Number of obs: 240, groups:  province_cfmg, 75; province, 8

Fixed effects:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                       3.249784   1.548580   2.099   0.0359 *
interviewee_rolecfmg_exec_member -1.205163   1.127959  -1.068   0.2853
interviewee_roletrad_leader      -1.578967   1.089285  -1.450   0.1472
interviewee_roledfo              -2.155322   1.122082  -1.921   0.0548 .
interviewee_roleNGO_rep          -1.588076   1.201137  -1.322   0.1861
interviewee_gendermale            0.790989   0.479019   1.651   0.0987 .
interviewee_age                  -0.009331   0.019952  -0.468   0.6400
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.687
```

```
intrvw_rlt_ -0.586  0.845
intrvw_rldf -0.786  0.858  0.854
intrvw_NGO_ -0.739  0.802  0.796  0.838
intrvw_gndr -0.171 -0.019 -0.082 -0.049    0.008
interview_g -0.714  0.125 -0.027  0.268    0.239 -0.036
optimizer (Nelder_Mead) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.00605841 (tol = 0.002, component 1)
```

```
 contrast                        odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member      3.337 3.764 Inf    1   1.068  0.8228
 cfmg_member / trad_leader           4.850 5.283 Inf    1   1.450  0.5954
 cfmg_member / dfo                   8.631 9.684 Inf    1   1.921  0.3061
 cfmg_member / NGO_rep               4.894 5.879 Inf    1   1.322  0.6774
 cfmg_exec_member / trad_leader      1.453 0.899 Inf    1   0.604  0.9745
 cfmg_exec_member / dfo              2.586 1.551 Inf    1   1.585  0.5073
 cfmg_exec_member / NGO_rep          1.467 1.078 Inf    1   0.521  0.9853
 trad_leader / dfo                   1.780 1.065 Inf    1   0.963  0.8717
 trad_leader / NGO_rep               1.009 0.747 Inf    1   0.012  1.0000
 dfo / NGO_rep                       0.567 0.377 Inf    1  -0.853  0.9138

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

We also examined which mechanisms are being used to achieve grievance and redress among the community female, male, and youth FGDs (Figure 12).

```r
prefix_griev <- "equity_and_inclusion_grievance_redress_mechanisms_"


griev_labels <- c(
  "community_meetings_or_forums_for_conflict_resolution"  = "Community meetings/forums",
  "mediation_by_traditional_leaders"                      = "Mediation by traditional leaders
  "formal_grievance_committees_within_the_cfm_group"      = "Formal grievance committees",
  "legal_channels_local_courts_government_agencies"       = "Legal channels",
  "external_mediation_by_ngos_or_third_parties"           = "External mediation (NGOs)",
  "anonymous_reporting_mechanisms_suggestion_boxes"        = "Anonymous reporting",
  "stipulated_in_constitution"                            = "Stipulated in constitution",
  "stipulated_in_redress_document"                        = "Stipulated in redress document"
)

full_griev_vars <- paste0(prefix_griev, names(griev_labels))
```

```r
cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  mutate(group = factor(
    case_match(
      interview_type,
      "cfmg_female" ~ "Female",
      "cfmg_male"   ~ "Male",
      "cfmg_youth"  ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female")
  )) |>
  select(group, all_of(full_griev_vars)) |>
  pivot_longer(all_of(full_griev_vars), names_to = "variable", values_to = "val") |>
  filter(val == 1) |>
  mutate(mechanism = factor(
    griev_labels[sub(prefix_griev, "", variable, fixed = TRUE)],
    levels = rev(griev_labels)
  )) |>
  count(group, mechanism) |>
  complete(group, mechanism, fill = list(n = 0)) |>
  ggplot(aes(y = mechanism, x = n, fill = group)) +
  geom_col(position = position_dodge(width = 0.75), width = 0.65, colour = "white") +
  geom_text(
    aes(label = ifelse(n > 0, n, "")),
    position = position_dodge(width = 0.75),
    hjust = -0.3, size = 3.5, colour = "grey30"
  ) +
  scale_fill_manual(
    values = c(
      "Female" = "#2A9D8F",
      "Male"   = "#57B8B0",
      "Youth"  = "#E9C46A"
    ),
    guide = guide_legend(reverse = TRUE)
  ) +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.15)),
    name   = "Number of FGDs"
  ) +
  scale_y_discrete(labels = \(x) str_wrap(x, width = 25), expand = expansion(add = 0.6)) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 14) +
```

```r
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.x        = element_text(colour = "grey20"),
  axis.text.y        = element_text(size = 14, colour = "grey20"),
  legend.position    = "bottom",
  plot.margin        = margin(10, 25, 10, 10)
)
```

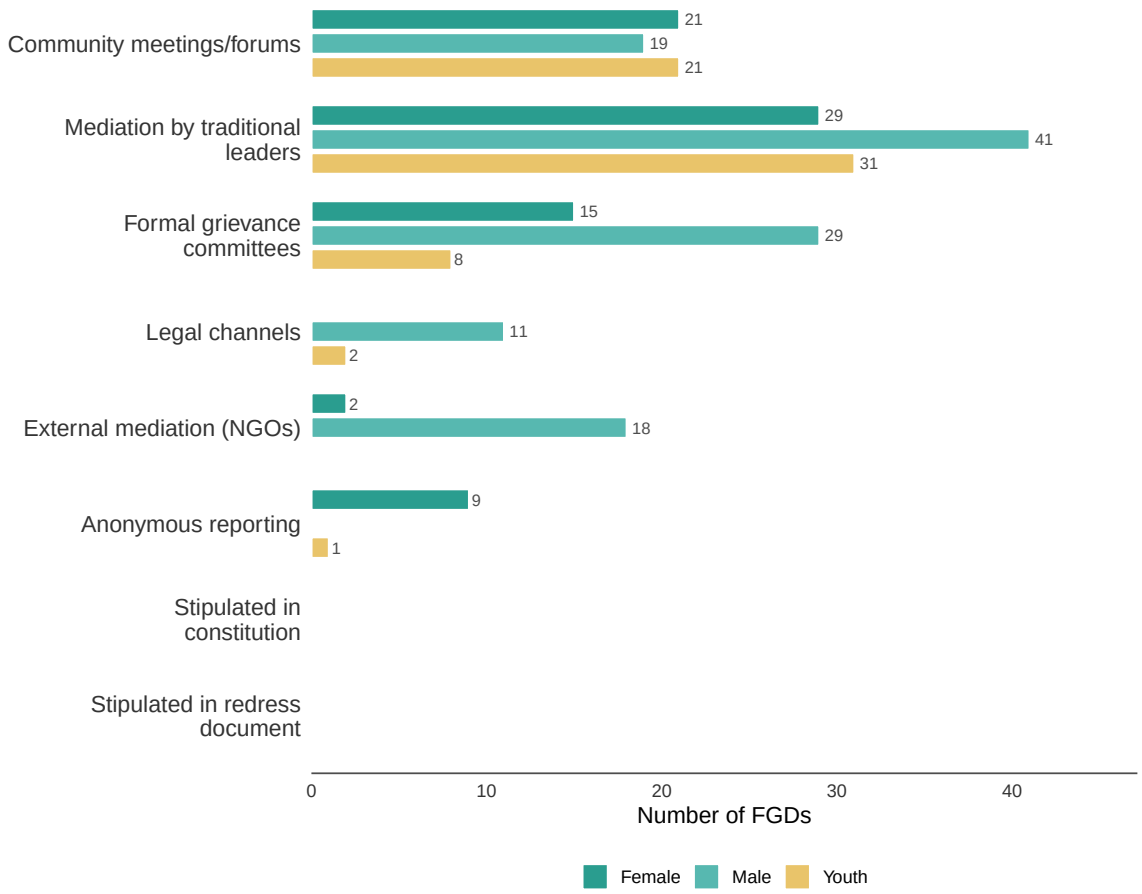

Figure 12: Mechanisms used for grievance and redress in CFMGs, as reported by community female, male, and youth FGDs. Bars show the number of FGDs reporting each mechanism.

Impact on poverty

We asked the community female, male, and youth FGDs to identify the ways in which CFM has had an impact on poverty in their communities (Figure 13).

```
prefix_pov <- "equity_and_inclusion_impact_on_poverty_"

pov_labels <- c(
  "increase_in_household_incomes"                                                    = "Incre
  "improved_nutrition_and_food_security"                                             = "Impro
  "improved_water_security"                                                          = "Impro
  "higher_school_attendance_rates"                                                   = "Highe
  "improved_access_to_healthcare"                                                    = "Impro
  "increase_in_household_assets_livestock_tools_housing_improvements"               = "Increa
  "creation_of_new_income_generating_activities_eco_tourism_handicrafts_ntfp_trade" = "New in
  "reduction_in_economic_inequality_within_the_community"                           = "Reduct
)

full_pov_vars <- paste0(prefix_pov, names(pov_labels))

cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  mutate(group = factor(
    case_match(
      interview_type,
      "cfmg_female" ~ "Female",
      "cfmg_male"   ~ "Male",
      "cfmg_youth"  ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female")
  )) |>
  select(group, all_of(full_pov_vars)) |>
  pivot_longer(all_of(full_pov_vars), names_to = "variable", values_to = "val") |>
  filter(val == 1) |>
  mutate(impact = factor(
    pov_labels[sub(prefix_pov, "", variable, fixed = TRUE)],
    levels = rev(pov_labels)
  )) |>
  count(group, impact) |>
  complete(group, impact, fill = list(n = 0)) |>
  ggplot(aes(y = impact, x = n, fill = group)) +
  geom_col(position = position_dodge(width = 0.75), width = 0.65, colour = "white") +
  geom_text(
```

```r
    aes(label = ifelse(n > 0, n, "")),
    position = position_dodge(width = 0.75),
    hjust = -0.3, size = 3.5, colour = "grey30"
  ) +
  scale_fill_manual(
    values = c(
      "Female" = "#2A9D8F",
      "Male"   = "#57B8B0",
      "Youth"  = "#E9C46A"
    ),
    guide = guide_legend(reverse = TRUE)
  ) +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.15)),
    name   = "Number of FGDs"
  ) +
  scale_y_discrete(labels = \(x) str_wrap(x, width = 25), expand = expansion(add = 0.6)) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.x        = element_text(colour = "grey20"),
    axis.text.y        = element_text(size = 14, colour = "grey20"),
    legend.position    = "bottom",
    plot.margin        = margin(10, 25, 10, 10)
  )
```

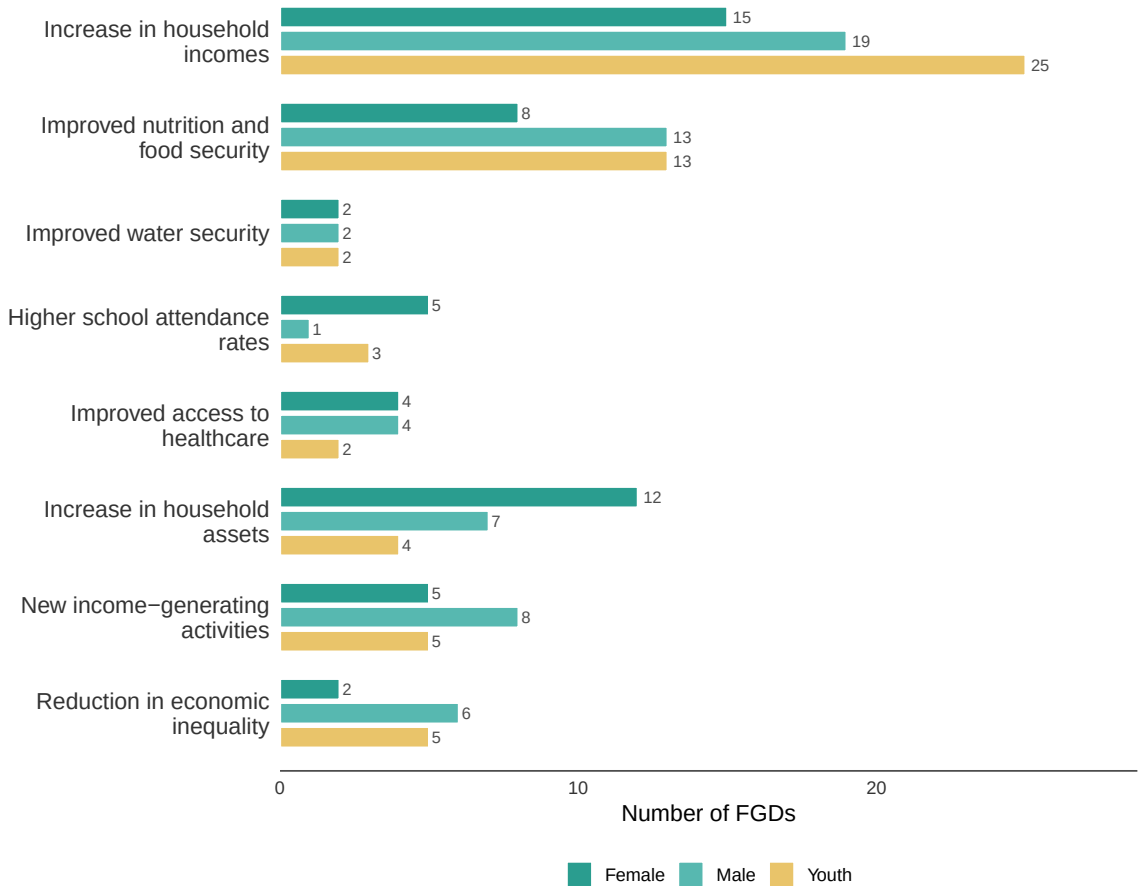

Figure 13: Reported impacts of CFM on poverty, as identified by community female, male, and youth FGDs. Bars show the number of FGDs reporting each impact.

Barriers to poorer community members

We asked the community female, male, and youth FGDs to identify the barriers that prevent poorer community members from benefiting from CFM (Figure 14).

```
prefix_b <- "equity_and_inclusion_challenges_for_poorer_community_members_"

barrier_labels <- c(
  "lack_or_limited_cfm_benefits"                                          = "L
  "limited_access_to_information_about_cfm_opportunities_and_benefits"    = "L
  "lack_of_financial_and_other_resources_to_participate_in_cfm_activities" = "La
  "exclusion_from_decision_making_processes_within_the_cfm_group"         = "Ex
```

```r
  "unequal_distribution_of_benefits_among_community_members"                    = "U
  "limited_access_to_training_or_capacity_building_opportunities"                = "Li
  "high_opportunity_costs_time_labor_or_resources_lost_by_participating_in_cfm"   = "Hig
  "social_or_cultural_barriers_discrimination_based_on_gender_age_or_status"     = "Soci
  "physical_distance_from_cfm_activities_or_resources"                           = "Ph
  "lack_of_trust_in_cfm_leadership_management_or_benefit_sharing_mechanisms"     = "No t
  "lack_of_interest_in_cfm_activities"                                           = "La
  "cfma_access_restrictions"                                                     = "C
  "influence_by_stronger_indiv"                                                  = "S
  "human_wildlife_conflicts"                                                     = "H
  "selfishness"                                                                  = "S
  "interference_from_govt_and_traditional_leadership"                            = "Go
)

full_b_vars <- paste0(prefix_b, names(barrier_labels))

# Frequency counts from binary checkbox variables
barrier_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_b_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = barrier_labels[sub(prefix_b, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

# Word frequencies from open-ended comments
comment_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(comment = equity_and_inclusion_challenges_for_poorer_community_members_comment) |>
  drop_na(comment) |>
  unnest_tokens(word, comment) |>
  anti_join(stop_words, by = "word") |>
  filter(nchar(word) > 2, !word %in% c(
    "cfm", "cfmg", "forest", "limited", "lack",
    "youths", "visible", "accessing", "people", "poorer", "community", "coming"
  )) |>
  count(word, name = "freq") |>
  filter(freq > 1)

# Combine checkbox labels and comment words
barriers_all <- bind_rows(barrier_freq, comment_freq) |>
```

```r
  group_by(word) |>
  summarise(freq = sum(freq), .groups = "drop")

set.seed(42)
ggplot(barriers_all, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
  scale_colour_gradient(low = "#F4A582", high = "#8B0000") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 14: Barriers preventing poorer community members from benefiting from CFM, as reported by community female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each barrier.

Addressing challenges

We asked the community female, male, and youth FGDs to suggest solutions to address the barriers facing poorer community members in benefiting from CFM (Figure 15).

```
prefix_sol <- "equity_and_inclusioNAddressing_challenges_"

sol_labels <- c(
  "provision_of_financial_support_or_subsidies_to_poorer_households"      = "Financial suppo
  "creation_of_targeted_programs_for_marginalized_groups"                  = "Targeted progra
  "community_meetings_or_forums_to_discuss_and_resolve_issues"             = "Community meeti
  "adjustments_to_benefit_sharing_arrangements_to_ensure_fairness"         = "Adjustments to
  "training_and_capacity_building_for_disadvantaged_groups"                = "Training and ca
  "leadership_and_governance"                                               = "Leadership and
)

full_sol_vars <- paste0(prefix_sol, names(sol_labels))

cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  mutate(group = factor(
    case_match(
      interview_type,
      "cfmg_female" ~ "Female",
      "cfmg_male"   ~ "Male",
      "cfmg_youth"  ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female")
  )) |>
  select(group, all_of(full_sol_vars)) |>
  pivot_longer(all_of(full_sol_vars), names_to = "variable", values_to = "val") |>
  filter(val == 1) |>
  mutate(solution = factor(
    sol_labels[sub(prefix_sol, "", variable, fixed = TRUE)],
    levels = rev(sol_labels)
  )) |>
  count(group, solution) |>
  complete(group, solution, fill = list(n = 0)) |>
  ggplot(aes(y = solution, x = n, fill = group)) +
  geom_col(position = position_dodge(width = 0.75), width = 0.65, colour = "white") +
  geom_text(
    aes(label = ifelse(n > 0, n, "")),
    position = position_dodge(width = 0.75),
```

```r
    hjust = -0.3, size = 3.5, colour = "grey30"
  ) +
  scale_fill_manual(
    values = c(
      "Female" = "#2A9D8F",
      "Male"   = "#57B8B0",
      "Youth"  = "#E9C46A"
    ),
    guide = guide_legend(reverse = TRUE)
  ) +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.15)),
    name   = "Number of FGDs"
  ) +
  scale_y_discrete(labels = \(x) str_wrap(x, width = 25), expand = expansion(add = 0.6)) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.x        = element_text(colour = "grey20"),
    axis.text.y        = element_text(size = 14, colour = "grey20"),
    legend.position    = "bottom",
    plot.margin        = margin(10, 25, 10, 10)
  )
```

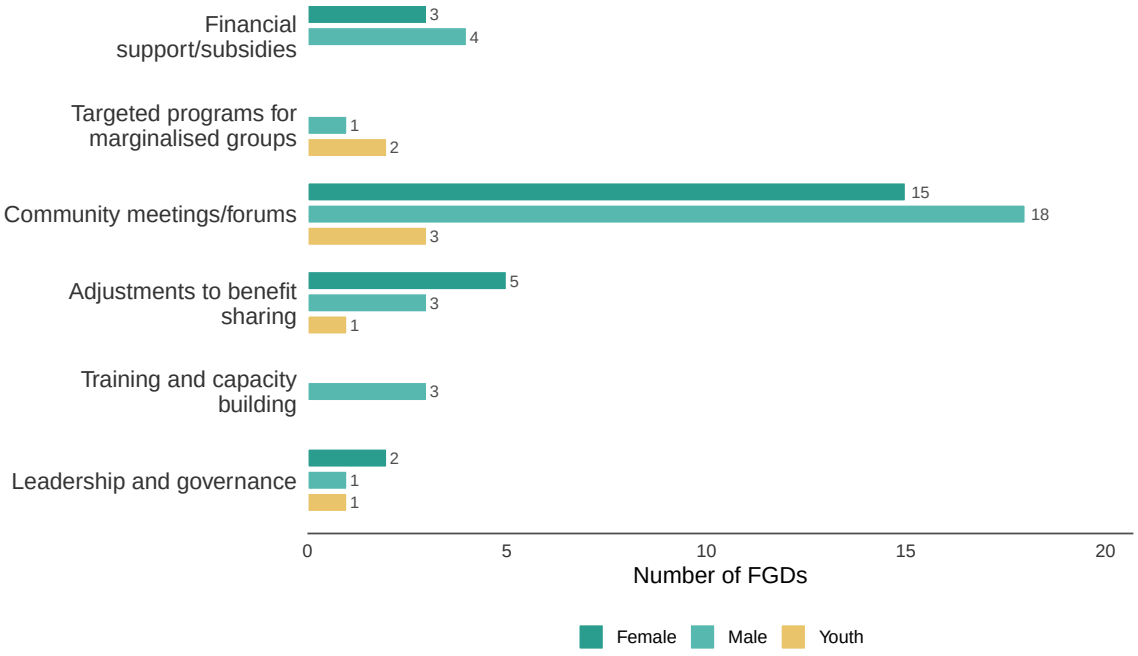

Figure 15: Suggested solutions to address barriers preventing poorer community members from benefiting from CFM, as reported by community female, male, and youth FGDs. Bars show the number of FGDs reporting each solution.